



Particle-resolved CFD-DEM with OpenHFDIB-DEM

Hands-on 1: Body Addition

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21st OpenFOAM Workshop

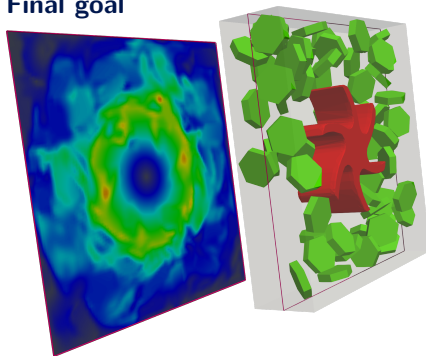


Overall course outline

OpenHFDIB-DEM – particle-resolved CFD-DEM



Final goal



General Info

Body Addition

Movement Modes

DEM

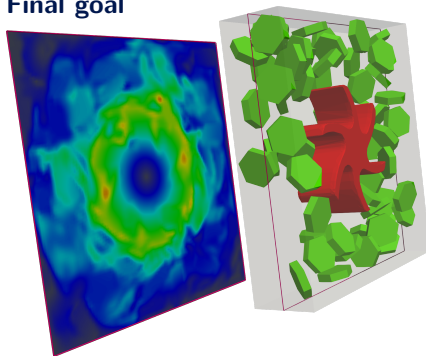
CFD-DEM

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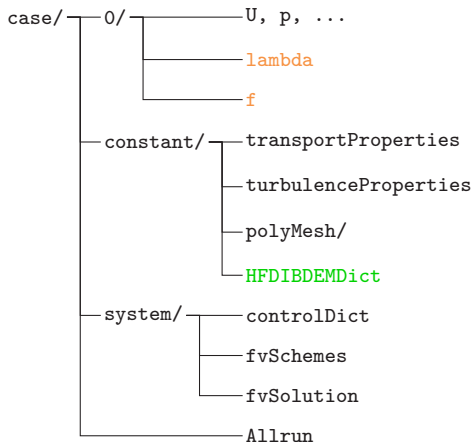
Movement Modes

DEM

CFD-DEM

Where are we in the workflow

The parts relevant to tutorial are in green

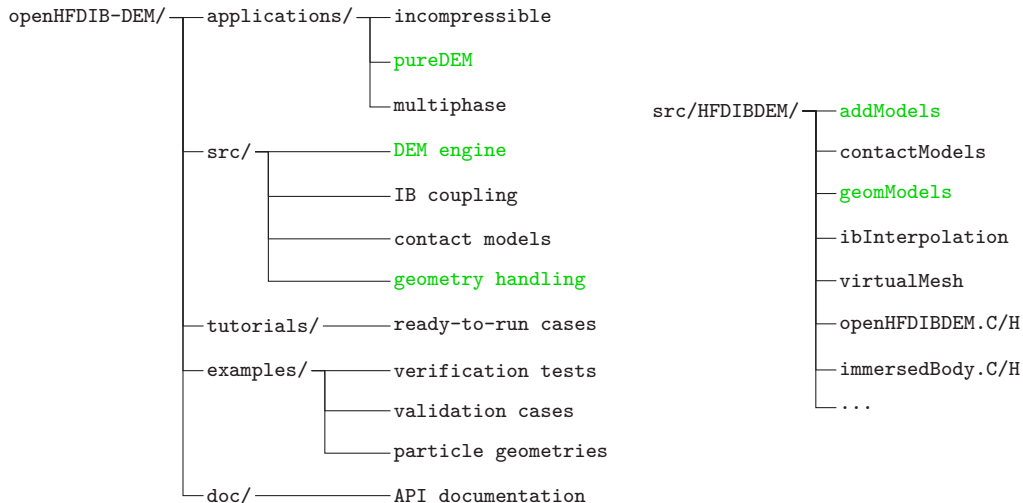


HFDIBDEMDict

- interpolation schemes for immersed boundary
- solid bodies definitions
 - body names
 - body types (sphere, convex, non-convex)
 - body motion (static, prescribed translation/rotation, fully coupled with fluid)
 - body addition models (addModels definition)
- DEM settings
 - DEM time step
 - solid materials
 - collision patches
 - contact treatment (virtual mesh)
- library output controls

Where to look if you are interested in the code

The parts relevant to tutorial are in green





Relevant Capabilities

Adding Single Body

Repeated Body Addition

Glitters and Impeller

—scan me—



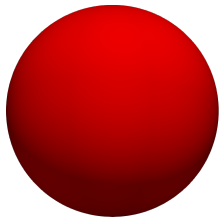
Geometric models in OpenHFDIB-DEM

Shape-based/spheres + STL-based/convex & non-convex



Shape-based

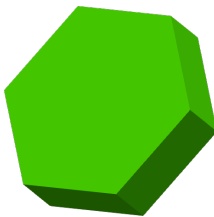
Sphere



```
1 bodyGeom sphere;  
2 sphere  
3 {  
4   radius 0.01;  
5   startPosition (0 0 0);  
6 }  
7
```

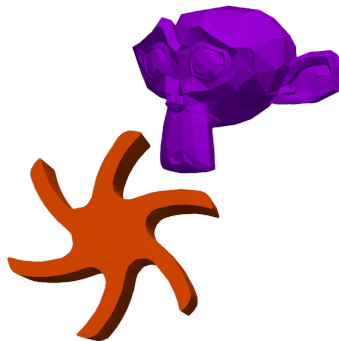
STL-based

Convex body



```
1 bodyGeom convex;
```

Non-convex body



```
1 bodyGeom nonConvex;
```

Note: for STL-based, position and size are property of STL

Simple body addition models in OpenHFDIB-DEM

Only once/Repeated/From file



Add Body Once

```
1 bodyAddition
2 {
3   addModel once;
4   onceCoeffs {};
5 }
6
```

Repeat Same Position

```
1 bodyAddition
2 {
3   addModel repeatSamePosition;
4   repeatSamePositionCoeffs
5   {
6     useNTimes 5;
7     timeBetweenUsage 0.01;
8   }
9 }
10
```

Add Once From File

```
1 bodyAddition
2 {
3   addModel onceFromFile;
4   onceFromFileCoeffs
5   {
6     fileName "path.txt"
7   }
8 }
9
```

constant/path.txt has to contain:

```
dir/0.stl
dir/1.stl
:
dir/100.stl
```

where dir:
constant/triSurface/dir

Advanced body addition models in OpenHFDIB-DEM



Random solids scattering $\rightsquigarrow$ Repeat Random Position

```
1 bodyAddition
2 {
3   addModel repeatRandomPosition;
4   repeatRandomPositionCoeffs
5   {
6     addMode fieldBased;//will be driven by solid volume fraction
7     fieldBasedCoeffs
8     {
9       fieldName lambda;
10      fieldValue 0.05;//target volume fraction in boundBox
11    }
12    addDomain boundBox;//will be added in a subdomain defined by boundBox
13    boundBoxCoeffs
14    {
15      minBound (xmin ymin zmin);
16      maxBound (xmax ymax zmax);
17    }
18    scalingMode noScaling;//all the solids will have the same size
19    noScalingCoeffs{}
20    rotationMode fixedAxisRandomRotation;//random rotation around the z-axis
21    fixedAxisRandomRotationCoeffs
22    {
23      axis (0 0 1);
24    }
25  }
26 }
```



Relevant Capabilities

Adding Single Body

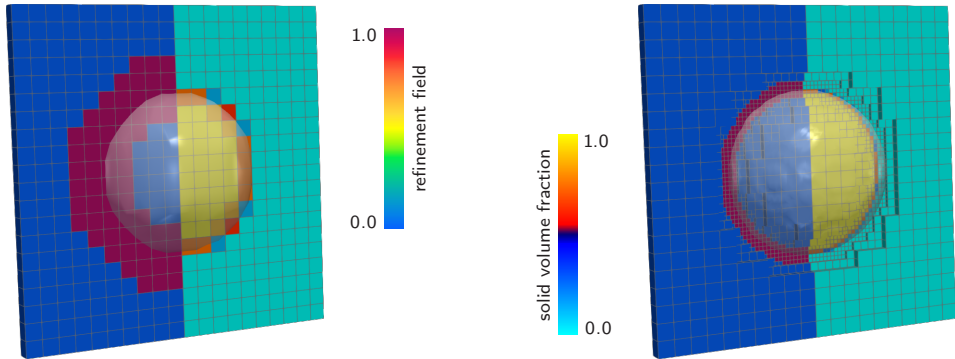
Repeated Body Addition

Glitters and Impeller

—scan me—



–test case presentation–





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Adding Single Body

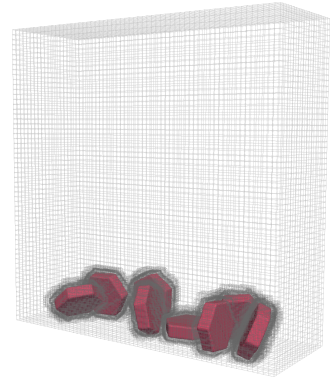
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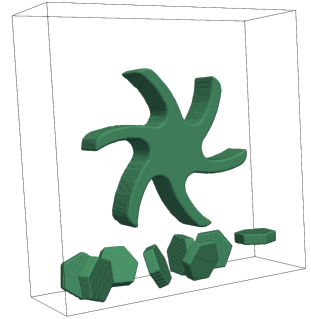
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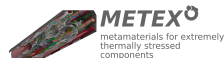


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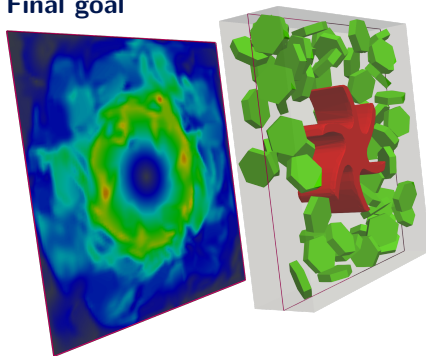


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